Lab 8

Logic Regression

```
LOGISTIC REGRESSION

                           Logit Regression Results
==============================================================================
Dep. Variable:                    low   No. Observations:                  189
Model:                          Logit   Df Residuals:                      185
Method:                           MLE   Df Model:                            3
Date:                Sun, 25 Oct 2026   Pseudo R-squ.:                 0.06250
Time:                        23:31:42   Log-Likelihood:                -110.00
converged:                       True   LL-Null:                       -117.34
Covariance Type:            nonrobust   LLR p-value:                  0.002125
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -1.3551      0.241     -5.611      0.000      -1.828      -0.882
smoke          0.6831      0.329      2.075      0.038       0.038       1.329
ht             1.4174      0.624      2.273      0.023       0.195       2.640
ui             1.0384      0.428      2.426      0.015       0.200       1.877
==============================================================================

Note: Classification metrics use training data.
Accuracy: 0.7037

Confusion Matrix (rows = actual; columns = predicted):
[[119  11]
 [ 45  14]]

Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.92      0.81       130
           1       0.56      0.24      0.33        59

    accuracy                           0.70       189
   macro avg       0.64      0.58      0.57       189
weighted avg       0.67      0.70      0.66       189

McFadden Pseudo-R-squared: 0.0625
```

Dataset: birthwt.csv

Predictors: smoke, ht, ui

Response Variable: low

1. What is the model's accuracy?

The logistic regression model has an accuracy of 70.37%. This means that the model correctly classified approximately 70% of the 189 observations based on the predictors smoking, hypertension, and uterine irritability.

2. What is the model's lowest precision?

The model's lowest precision is 0.56, or 56%, for category 1 (low birthweight). This means that 56% of the observations predicted to have low birthweight were correctly classified.

3. Do all categories have at least 30 support?

Yes, both categories have at least 30 observations. Category 0 has a support of 130, while category 1 has a support of 59. Therefore, both categories meet the minimum support requirement.

4. Based on the pseudo-R^2 value, is this model good in this scenario?

The model has a pseudo-R^2 value of 0.0625, which is relatively low. This suggests that the predictors provide only a modest improvement over a model without predictors. Although the model has an accuracy of 70.37%, its low pseudo-R^2 and poor recall for low birthweight cases indicate that it is not particularly strong for predicting low birthweight.

Count Variable Analysis

```
POISSON REGRESSION

WARNING: Response includes fractional values. Standard Poisson regression assumes integer counts. Results are
exploratory.

                    Poisson Regression Results
==============================================================================
Dep. Variable:       mhlth_crudeprev   No. Observations:              2299
Model:                      Poisson   Df Residuals:                  2287
Method:                        MLE   Df Model:                        11
Date:             Sun, 25 Oct 2026   Pseudo R-squ.:               0.04356
Time:                     23:34:36   Log-Likelihood:              -5397.6
converged:                    True   LL-Null:                     -5643.4
Covariance Type:          nonrobust   LLR p-value:                1.908e-98
==============================================================================
=
```

```
                       coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------
const                  1.7032      0.172      9.888      0.000       1.366       2.041
binge_crudeprev        0.0032      0.003      0.954      0.340      -0.003       0.010
checkup_crudeprev     -0.0004      0.002     -0.192      0.848      -0.004       0.004
csmoking_crudeprev     0.0043      0.004      1.100      0.271      -0.003       0.012
depression_crudeprev   0.0137      0.002      5.800      0.000       0.009       0.018
obesity_crudeprev      0.0001      0.002      0.076      0.940      -0.003       0.004
sleep_crudeprev        0.0024      0.002      1.059      0.289      -0.002       0.007
selfcare_crudeprev    -0.0608      0.022     -2.708      0.007      -0.105      -0.017
indeplive_crudeprev    0.0522      0.014      3.759      0.000       0.025       0.079
loneliness_crudeprev   0.0113      0.003      3.717      0.000       0.005       0.017
foodinsecu_crudeprev  -0.0083      0.007     -1.247      0.212      -0.021       0.005
housinsecu_crudeprev   0.0150      0.008      1.786      0.074      -0.001       0.032
==============================================================================================
=

Variance Inflation Factors (VIF):
          Predictor    VIF
    binge_crudeprev  2.529
   checkup_crudeprev  1.750
  csmoking_crudeprev  6.935
depression_crudeprev  2.216
   obesity_crudeprev  2.573
    sleep_crudeprev  3.209
  selfcare_crudeprev 30.514
 indeplive_crudeprev 30.953
loneliness_crudeprev  3.433
foodinsecu_crudeprev 62.831
housinsecu_crudeprev 46.840

McFadden Pseudo-R-squared: 0.0436
```

Dataset: places_2025.csv

Response Variable: mhlth_crudeprev

1. Your model should include 4 significant predictors. What are they?

The four statistically significant predictors are depression_crudeprev, selfcare_crudeprev, indeplive_crudeprev, and loneliness_crudeprev. All four predictors have p-values below 0.05, indicating statistically significant relationships with the response variable.

2. Which of these predictors is the most significant (hint: look at the 'coef' column)?

Selfcare_crudeprev has the largest coefficient in absolute value at -0.0608. This indicates a negative relationship with the response variable. Among the four significant predictors, self-care has the strongest estimated coefficient per one-unit increase, although coefficient size alone does not establish statistical significance.

3. Do any of your significant predictors have an overinflated VIF (VIF > 10)?

Yes, two significant predictors have VIF values greater than 10. Selfcare_crudeprev has a VIF values greater than 10. Selfcare_crudeprev has a VIF of 30.514, and indeplive_crudeprev has a VIF of 30.953. These values indicate substantial multicollinearity, which can make the model's coefficient estimates less reliable.

4. How much of the variability is captured by the model (hint: look at pseudo-R^2)?

The model has a pseudo-R^2 value of 0.0436, or approximately 4.36%. This indicates a relatively small improvement in model fit compared to the null model. Unlike traditional R^2, pseudo-R^2 does not directly represent the percentage of variability explained. Therefore, the model has limited explanatory power.